



7th Global AI Hackathon

C H A L L E N G E

05

.....

AI Atlas for the World's Rare Diseases

Connect the world's knowledge so patient groups facing an untreated rare disease can find a path forward

S U P P O R T E D B Y



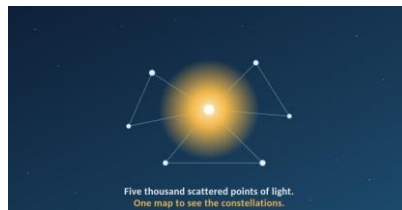
Goals and Motivation

Imagine learning that your child has a rare genetic disease—and that no approved treatment exists. You know the gene's name, but it does not tell you who could help or what to do next. Like **Maria**, the patient leader in this challenge, you begin finding other families, raising money, and asking researchers where to start. A diagnosis has made you the organizer of a research effort you never expected to lead.

10,000 known rare diseases	80% trace back to DNA	5,000 are monogenic	350M people affected worldwide	<5% have an approved treatment
--------------------------------------	---------------------------------	-------------------------------	--	---

THE 10× GOAL

Help rare disease research advance toward a possible treatment 10× faster.



Five thousand scattered points of light. One map to see the constellations.

Why now. Disease vocabularies and public research databases provide a foundation for connecting this information. Language models can help extract relationships from papers; graph analytics can reveal links across genes, symptoms, studies, and communities. NIH projects such as [RARE-SOURCE](#) already combine biomedical data and AI. The opportunity is to turn those capabilities into an understandable, evidence-backed route that patient groups can act on.

Current Challenges

KNOWLEDGE IS SCATTERED	Moving toward a treatment requires connected work: understanding the biology, gathering patient data, choosing experiments, and bringing the right researchers and clinical partners together. Yet the relevant knowledge sits across papers, disease databases, studies, and patient organizations.
GROUPS REBUILD WHAT EXISTS	Another community may already have a useful registry or experimental model. Maria cannot reuse what she cannot find—or judge whether it applies. Groups spend scarce time and funding rebuilding similar assets.
NAMES HIDE MECHANISMS	Different genes can disrupt the same biological process; one gene can have different effects; similar symptoms can have different causes. Organizing by disease name misses the research a community could share.
THE KEY INSIGHT	A monogenic disease can be caused by variants in one gene, but its research pathway may connect to many diseases. Organizing knowledge by mechanism and phenotype —the pattern of symptoms—may reveal opportunities to share research. Each proposed link needs evidence.

Your Challenge

Build the **AI Atlas for the World's Rare Diseases**: a knowledge graph whose nodes represent diseases, genes and variants, mechanisms, symptoms, patient groups, papers, studies, and research assets. Its edges explain how those things relate and cite the evidence. Use AI to assemble and reconcile the information, then graph analytics to find meaningful clusters. **Give Maria an interface that carries her from her disease to a supported connection, an existing asset, a collaborator, and a concrete next step toward treatment.**

Three questions drive the atlas. Maria needs to know: **Who shares our disease characteristics? What useful work already exists? What should we do together next?** The atlas should find the dots nobody's connected, explain *why* they're connected with cited evidence, and turn that into an action a patient group leader can take this week.

The People Who Must Turn Knowledge into Progress

Maria leads the journey. Devon brings the family's needs; Priya assesses therapeutic opportunities; Dr. Osei tests the science.

Maria <i>Patient organization leader</i>	Pain point: Maria leads a patient group for a disease with no approved treatment. She is assembling families, clinical observations, a registry, and research partners from scratch. She cannot tell whether a related group has already built an asset she could reuse or whether a seemingly similar disease follows a different mechanism. What changes: An explainable graph shows her closest disease clusters, the genes and symptoms connecting them, which evidence comes from papers or studies, and which patient groups are already working on the same pathway. It highlights a shared clinical opportunity and the next experiment needed to validate it. <i>Optional but valuable:</i> investors who've backed similar modalities before, and funders whose RFAs quietly signal where money is about to move. <i>"We may be the only family with this diagnosis, but this pathway connects us to another community and an existing study. Here is what must be checked before we join forces."</i> Leans on: Clustering · Pathway navigator · Connector
Devon <i>Newly diagnosed patient / caregiver</i>	Pain point: Days into a diagnosis, Devon runs 2 a.m. searches with zero medical background and zero community. Search engines return either nothing useful or dense journal abstracts written for specialists. What changes: Devon finds their exact community when one exists, or an evidence-qualified connection to related communities, research, and support. If no useful connection is known, the atlas says so clearly and shows what evidence could change that. <i>"Here is the patient group for your exact diagnosis. If none exists, here are the closest related communities and how to help build the missing one."</i> Leans on: Search engine
Priya <i>Biotech / pharma scout</i>	Pain point: Priya holds one therapeutic mechanism—a way to replace a missing protein, say, or silence a toxic one—and needs every disease cluster it could plausibly treat. Evaluating candidates one disease at a time, scattered across papers, conference talks, and cold outreach, takes months per mechanism. What changes: A ranked view of clusters with mechanistic evidence, patient communities, existing assets, and unmet needs gives Priya a concrete route to assess therapeutic potential. <i>"A ranked list of clusters matching your mechanism, each annotated with active patient advocacy groups, existing infrastructure, and named contacts."</i> Leans on: Clustering · Search engine
Dr. Osei <i>Academic researcher / clinician-scientist</i>	Pain point: Dr. Osei runs a lab devoted to one gene and one rare disease, publishing papers and applying for grants inside an academic silo. He has no way of knowing a colleague three universities away is chasing an "unrelated" disease that is, mechanistically, nearly the same problem. What changes: An evidence-backed cluster shows Dr. Osei where to test a shared mechanism, recruit collaborators, and reuse existing patient-group infrastructure. <i>"Who else works on my mechanism, across every gene name that mechanism can hide under—plus a way to reach them."</i> Leans on: Connector

Modules

Module 1: Connect the Evidence (Graph Builder)

Start with one disease cluster—then scale up to all of the world's medical information. Resolve names and synonyms, assign stable identifiers, and **record the source, date, and confidence of each relationship**. These connections let the atlas answer Maria's questions:

Edge	Source	What it does
Gene–variant–mechanism	OMIM, ClinVar	Connects molecular changes to their effects, rather than treating shared gene names as sufficient
Disease–phenotype	HPO	Identifies symptom patterns shared across genes and distinguishes broad symptoms from unusually informative ones
Publication–claim–investigator	PubMed / PMC	Traces a proposed relationship to its supporting or contradictory research and the people studying it

Edge	Source	What it does
Study–condition–intervention	ClinicalTrials.gov	Finds studies or designs that might inform a cluster; assess eligibility and mechanism before suggesting a shared trial
Funding–researcher–asset	NIH RePORTER	Shows current programs, reusable infrastructure, and gaps along the proposed clinical pathway
Patient organization–disease–registry	NORD, Global Genes, Orphanet, verified group sites	Reveals communities that could pool evidence and reduce duplicated work

Module 2: Make Each Connection Trustworthy

A promising graph path matters only if Maria and a scientist can inspect it. Show the supporting source, distinguish observations from inferred links, and surface contradictory findings. Help the group decide what to investigate next.

No supported route?

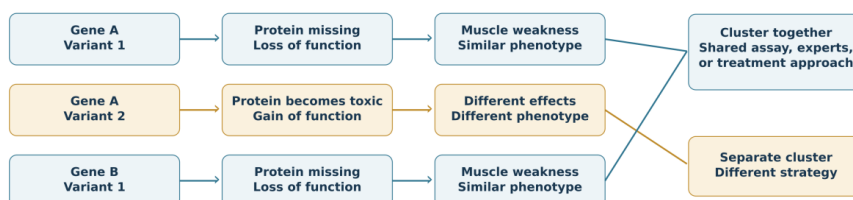
Say so. Explain the search coverage and the missing evidence, so the group knows what is unknown and what to test next.

Module 3: Turn the Network into Action

Use the graph to surface opportunities no single disease community could see alone:

1. Mechanistic overlap	Cluster diseases by variant effect, pathway, and phenotype rather than by name or category, so a lysosomal storage disease and a differently-named disease sharing the same pathway show up as neighbors.
2. Find what can be shared	Map registries, natural history studies, models, biomarkers, and clinical studies onto the cluster. Identify duplicate efforts and test whether assets or study designs could be adapted across communities.
3. Network overlap	Surface when two “unrelated” disease communities already share a key opinion leader (researcher, clinician, biotech rep, investor/funder).

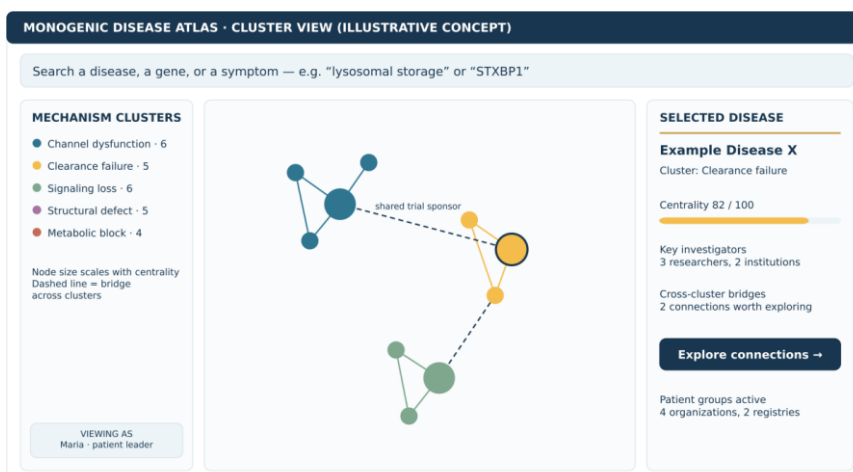
A gene name is only the starting point



Same gene, different mechanisms. Different genes, shared mechanisms and patient experience.

Make the Network Understandable to a Family

The graph is the engine; the experience must make its meaning clear. Someone facing an untreated disease should be able to see why a connection matters, what remains uncertain, and who could help. Use the concept below as a starting point and innovate on the experience.



Illustrative starting point: a disease map becomes valuable when it helps Maria understand a connection and act on it. Design your own way to make that journey clear.

Low ink, high signal	Whitespace does the design work; color carries meaning alone.
One global search	Disease, gene, symptom, patient group, or mechanism can open the same graph, with clear synonym resolution.
Progressive reveal	Summary first, depth on click.
Explain every edge	Put source, relationship type, confidence, and contradictory evidence beside the connection it supports.
Patient action view	Show shared assets, possible clinical pathways, partners, and next experiments; distinguish viable leads from unsupported links.

Stretch Goals

- **Invent new uses for the network.** Propose an experiment that tests a shared mechanism, identify an overlooked research partner, or let patient groups contribute missing evidence.
- **Reduce the burden of care.** Explore how these connections could help patients while treatment research progresses. Choose the opportunity you believe could change patients' lives most.

Think Bigger: The 10× Moonshot

Choose a meaningful milestone toward a possible treatment, such as launching a shared natural history study or reaching a decision on a therapeutic candidate. Compare the existing timeline with your proposed route and explain the assumptions behind a 10× acceleration.

YOUR 24-HOUR AMBITION	Demonstrate one complete journey from disease to connection to shared action. Make the case for how it could shorten a meaningful step toward treatment by 10×, and show what would need to be validated next.
------------------------------	--

Built with OpenAI

- **Extract.** Pull genes, variants, phenotypes, claims and investigators out of papers and abstracts into graph edges, each tied to its source.
- **Reconcile.** Resolve names and synonyms across disease vocabularies so one disease, gene or symptom maps to one stable node.
- **Explain.** Turn a graph path into plain language a family can follow, with every step citing the edge that supports it.

If you want to win the challenge track prizes, then you would need to leverage OpenAI's models or tools.

Data Sources and Hints

Begin with a focused slice that has enough biology, research, and patient-group information to demonstrate a complete path. Show how the graph can grow as new evidence and communities contribute.

Category	Source	Use
Patient-group directories	NORD , Global Genes , Orphanet , EURORDIS , Rare Disease UK , Genetic Alliance	Find communities, registries and the groups already working on a pathway
Research signal	PubMed / PMC , NIH RePORTER	Claims, investigators, funding and active programs
Biology engine	OMIM , HPO , MONDO , ClinVar	Genes, variants, phenotypes and stable disease identifiers
Assets already built	ClinicalTrials.gov , patient-org websites and press releases	Studies, registries and reusable designs
Stretch sources	Jackson Laboratory , RareConnect , bioRxiv , medRxiv	Models, communities and the newest preprints

What Good Looks Like

Maria types her disease into one search box. The atlas follows it to a disrupted pathway, another gene, a related disease, and the patient group working on it. Every edge has an explanation and a source. Next, she discovers an existing registry and study design: the atlas shows what is reusable, what differs between the diseases, and which biological or eligibility questions need expert review. Finally, she approaches a partner with a sourced proposal for shared research. If the graph reveals no supported lead, she leaves with a clear account of what is unknown and a next question to test. That is the bar.

What Makes a Strong Submission

Criterion	What judges look for
Graph quality	Meaningful node and edge design, defensible clustering, useful paths, counterexamples, and clear treatment of uncertainty
Evidence integrity	Sourced, cross-checked claims that distinguish data from hypotheses and clinical proof
Patient progress	A family or group moves from an isolated diagnosis to a justified collaboration, reusable asset, and next research or clinical milestone
10× impact	A meaningful milestone toward a possible treatment, the existing timeline versus your route, and the assumptions behind a 10× acceleration
Ambition and product craft	An intuitive experience that helps patient groups collaborate at a scale no single disease community could achieve alone

What to Submit

- **Working prototype.** Deployed or easy to run locally, ready for judges to search and explore live.
- **Source code repository.** With a README covering architecture and how to reproduce the dataset.
- **A Team video and a 1-minute walkthrough.** Follow a family or patient group through the graph to either a justified collaboration and next step, or an honest gap with a plan to investigate it.

Give patient groups the connections to turn an isolated diagnosis into a shared path toward treatment.

Build boldly. Make the evidence understandable. Show the next step.